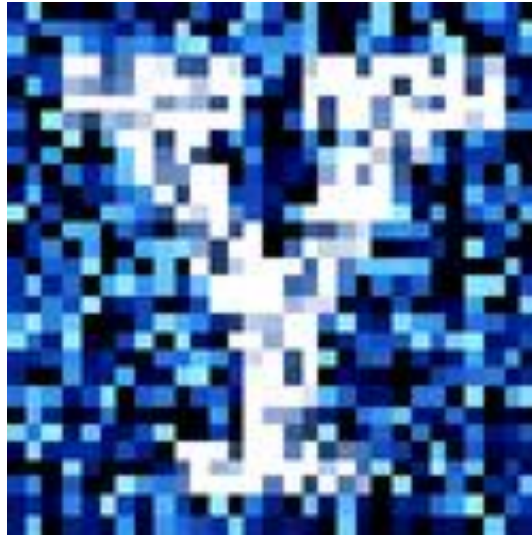


YData: Introduction to Data Science



Lecture 26: Data Ethics and Final Review

Overview

Ethics

Wrap-up

Project timeline

Friday, April 24th

- Project is due on Gradescope
 - Add peer reviews to an Appendix of your project
 - A late submission deadline is added to Sunday, Apr 26th

Please also fill out the final project reflection on Canvas!

- It will be very valuable to have your feedback on how the project and class overall went

Announcement

Exam review session: Tuesday April 28th by Shivam to review the practice final (details to TBD)

Final exam: Thursday April 30th at 2pm

- **Location:** Marsh Auditorium

Also, be sure to download any work from the class you want to save from the JupyterHub server since accounts will be deleted toward the beginning of next semester



Final exam logistics

- The exam is comprehensive.
- A practice final is posted.
- You are allowed an *additional* one-page (double-sided) cheat sheet that has the same rules as the midterm cheat sheet.
 - You can reuse the cheatsheet from the midterm.
 - This means you have two cheatsheets in total, one new + one from midterm.
 - You can get it back from my mailbox in SDS 11th floor of Kline. If your midterm cheatsheet is typed, you can simply reprint it too.
 - If you lost the midterm cheatsheet, it is okay to prepare a new one.

Ethics

Ethics in Data Science

Ethics of:

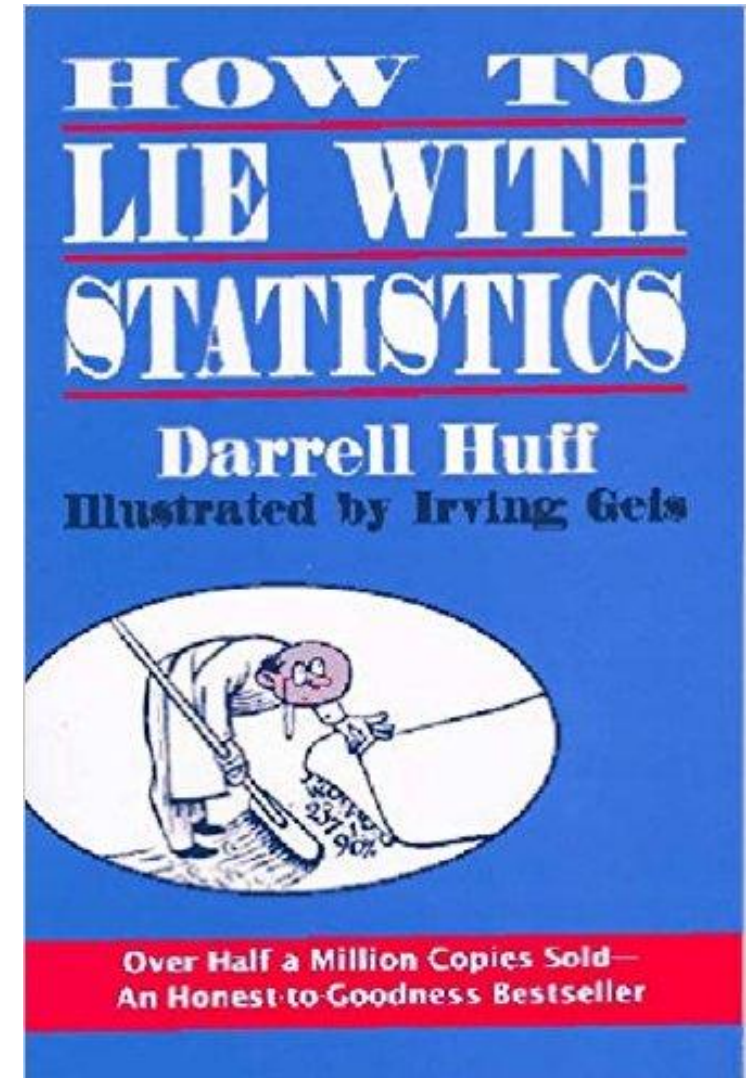
1. Data presentation
2. Using valid data
3. Data scraping TOS and privacy
4. Reproducibility
5. Ethics in Statistical analyses
6. Ethics of creating powerful tools

1. Ethics of data presentation

Data should be displayed in an honest way that gives an accurate picture of trends

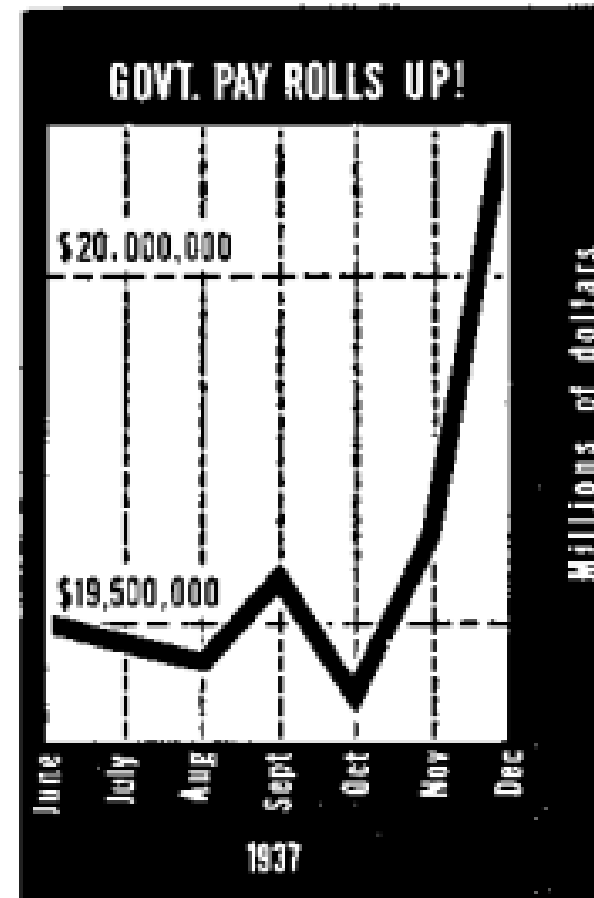
Darrell Huff wrote a classic book in the 1950's pointing out ways that people lie with statistics

The book was banned as training material at the VA



Ethics of data presentation

What is potentially misleading with this figure?



From a 1938 article in Dun's Review titled 'GOVERNMENT PAY ROLLS UP!'

Case in action in 2026

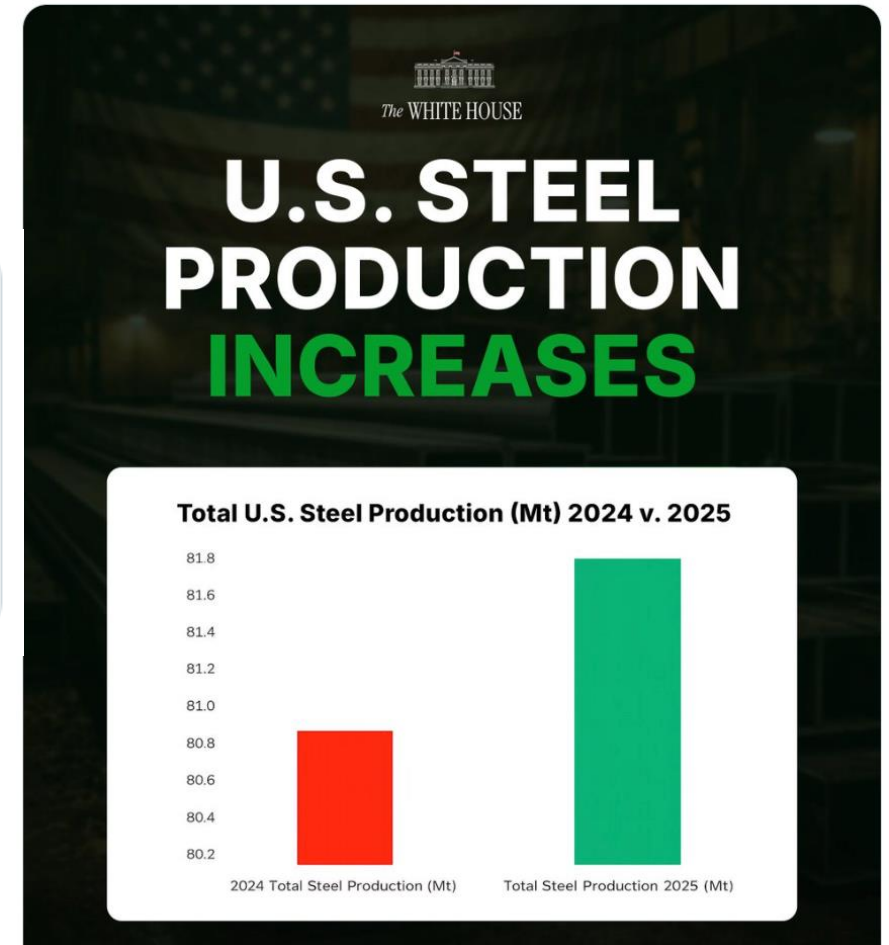
 Readers added context they thought people might want to know →

The vertical axis only shows values in the range 80.2Mt to 81.8Mt. This makes the 1% increase appear much larger than it actually is.

tradingeconomics.com/united-states/...

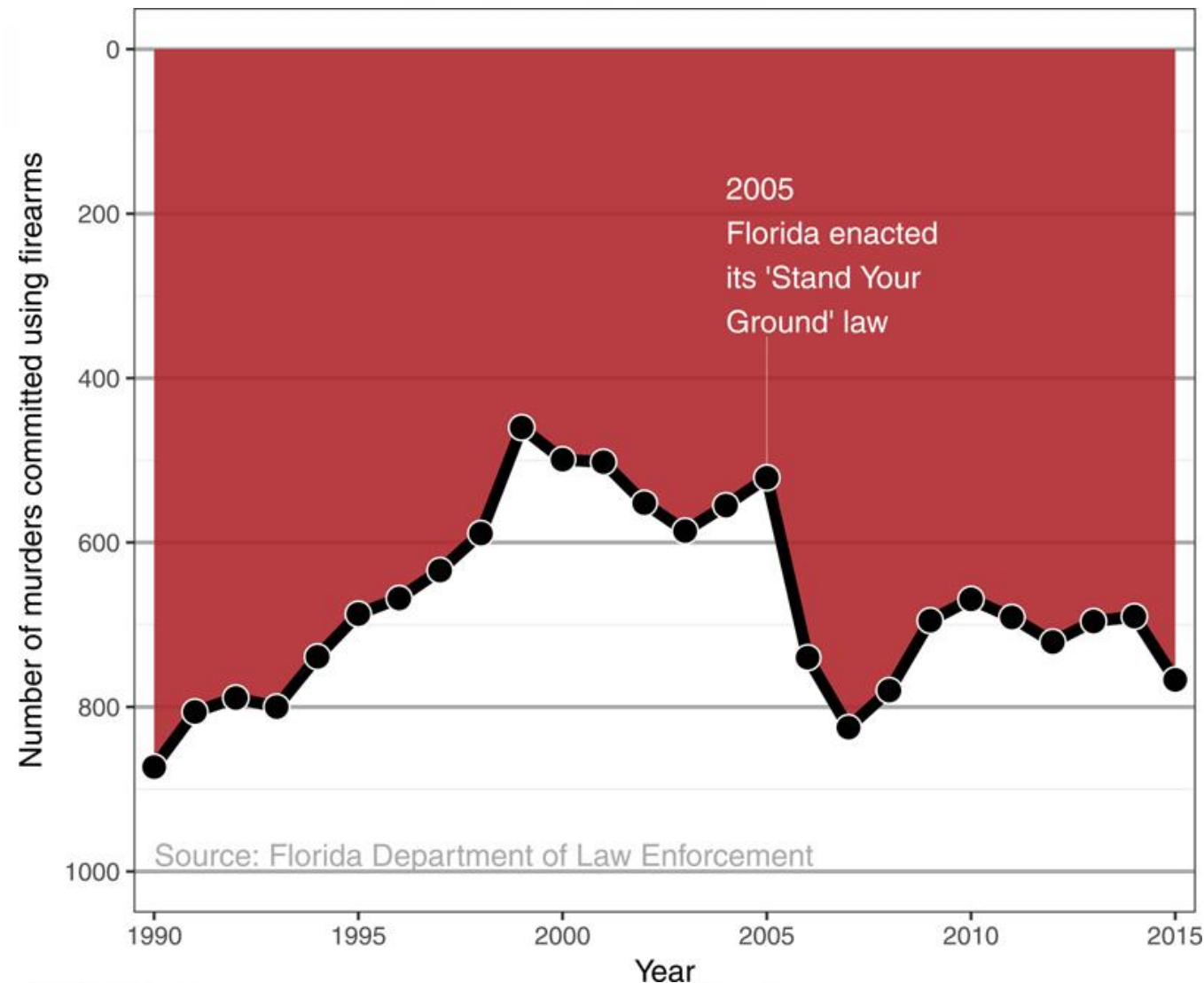


American steel is BACK. 🇺🇸



Did 'Stand Your Ground' decrease murder by firearms?

What is misleading with this figure?



2. Using valid data

Is almost everyone
satisfied with Hampshire
College?

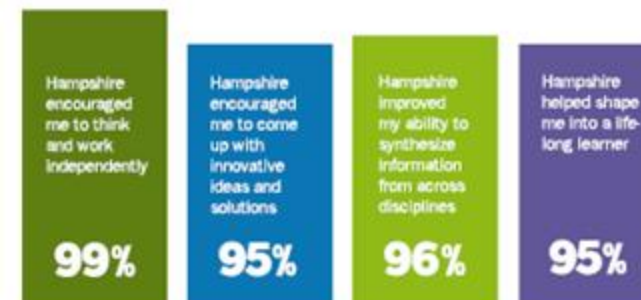
Alumni Survey Results

As part of a strategic-planning process, in spring 2013 Hampshire College launched a survey of alums. Via email, the College invited 8,160 alums to fill out an online questionnaire administered by the campus's Alumni and Family Relations and Institutional Research offices. A total of 1,920 surveys were completed, yielding a response rate of 24%.

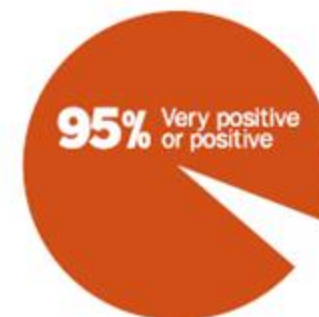
Note: The percentages in the data (below) are based on the number of responses received for each question.

To what extent do you agree with the following statements?

Strongly Agree or Agree



Please rate
your student
experience at
Hampshire.



3. Data scraping, terms of service and privacy

Scraping publicly available data is fine (e.g., Wikipedia) but what about scraping data if:

- It violates a website's Terms of Service?
- User privacy?

Kirkegaard and Bjerrekaer scraped okcupid and data on 68,371 users publicly available including usernames, dating preferences, etc.

- Is this ok?

Submitted: 8th of May 2016

Published: 3rd of November 2016

The OKCupid dataset: A very large public dataset of dating site users

Emil O. W. Kirkegaard*

Julius D. Bjerrekaer[†]



Open Differential
Psychology

4. Reproducibility

Do scientists have an ethical obligation to make sure their research is reproducible?

nature|methods

Access provided by Massachusetts Institute of Technology



Altmetric: 5

Citations: 5

[More detail >>](#)

Commentary

Ethical reproducibility: towards
transparent reporting in biomedical
research

Reproducibility

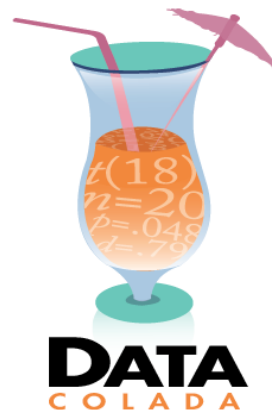
Do scientists have an obligation to share data/code?

- What if it could hurt your career?
 - Others could prove you wrong, make new findings on your own data, etc.

What should you do if you find one of your papers is wrong?

- You need to retract the paper!

Retraction Watch



NEWSLETTERS SIGN IN NPR SHOP

NEWS CULTURE MUSIC PODCASTS & SHOWS SEARCH

EDUCATION

Harvard professor who studies dishonesty is accused of falsifying data

JUNE 26, 2023 · 1:15 PM ET



Juliana Kim



Francesca Gino has been teaching at Harvard Business School for 13 years.
Maddie Meyer/Getty Images

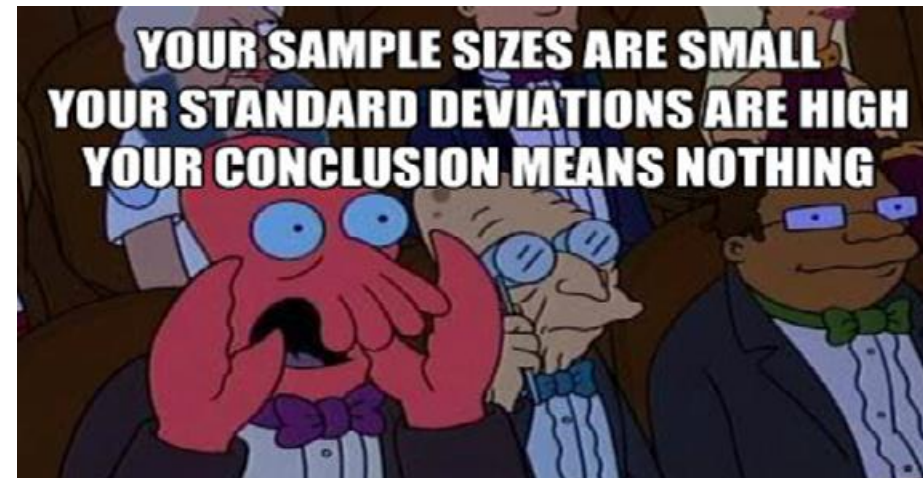
5. Ethics in Statistics

P-hacking (data dredging):

Keep trying different hypothesis tests on a data set until you reach 'statistical significance' ($p < 0.05$)

File drawer effect:

- Try a million studies until one is significant



6. Ethics of creating powerful tools

Some prominent people are concerned about job loss due to machine learning, or even computers posing an existential threat to humans

- Is this something we should be concerned with as Data Scientists?



Ethics in machine learning

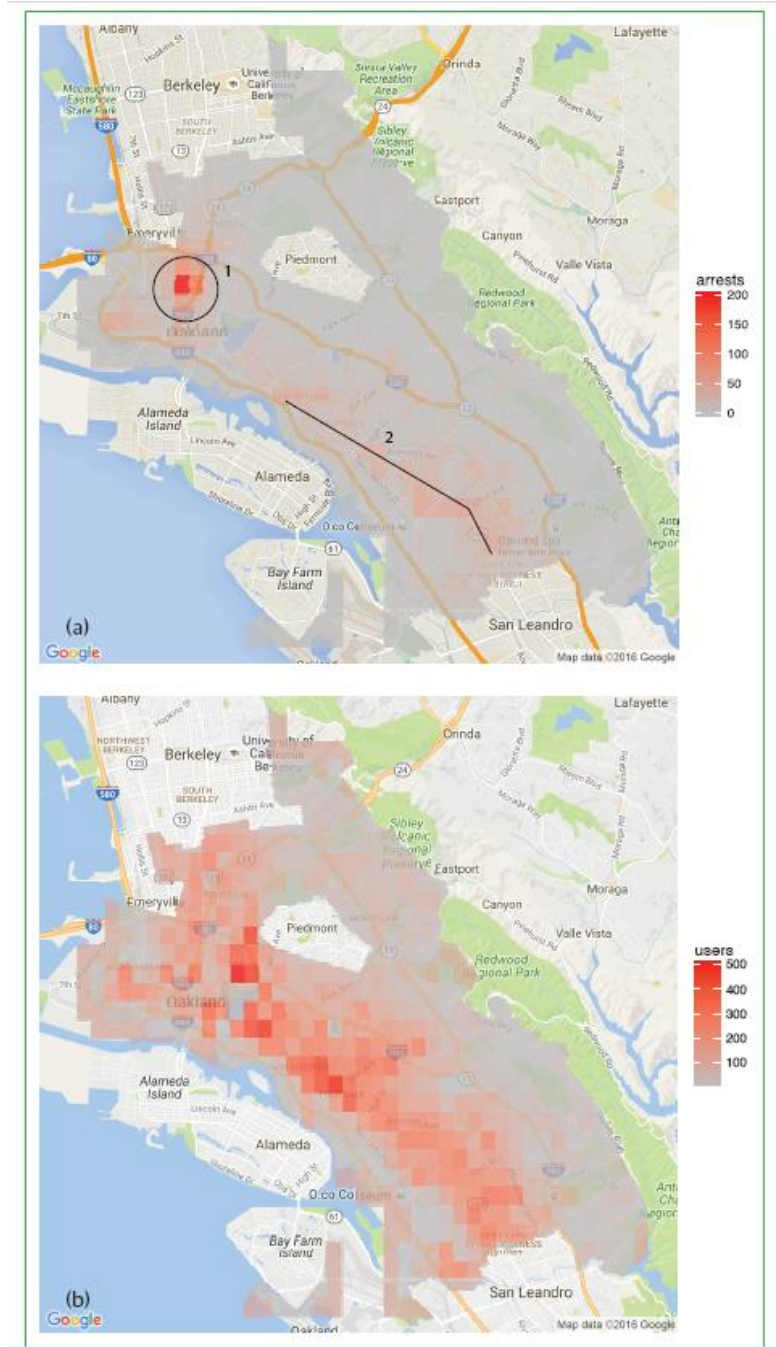
Idea: use ML to police areas with most crimes

- E.g. more police where most drug arrests have made in the past

Possible results

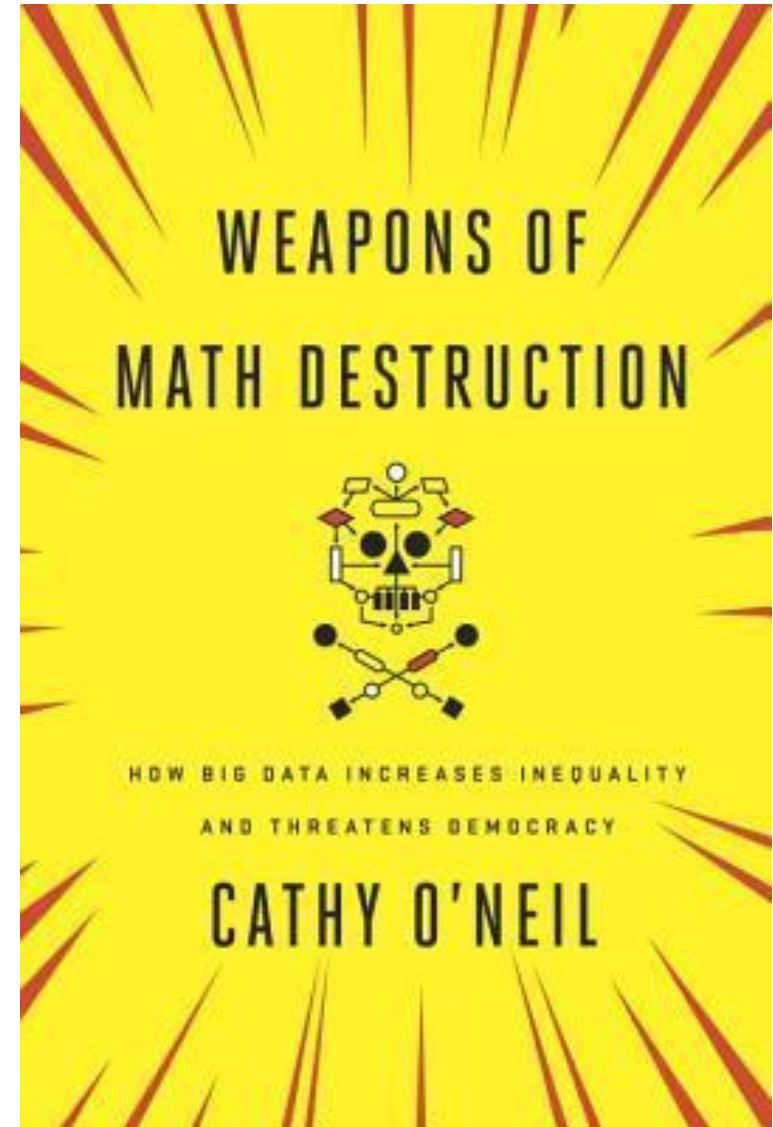
- Continued higher arrest rates in these areas seemingly showing the ML algorithm is working

Any potential problems with this?



Additional reading

https://www.ted.com/talks/cathy_o_neil_the_era_of_blind_faith_in_big_data_must_end

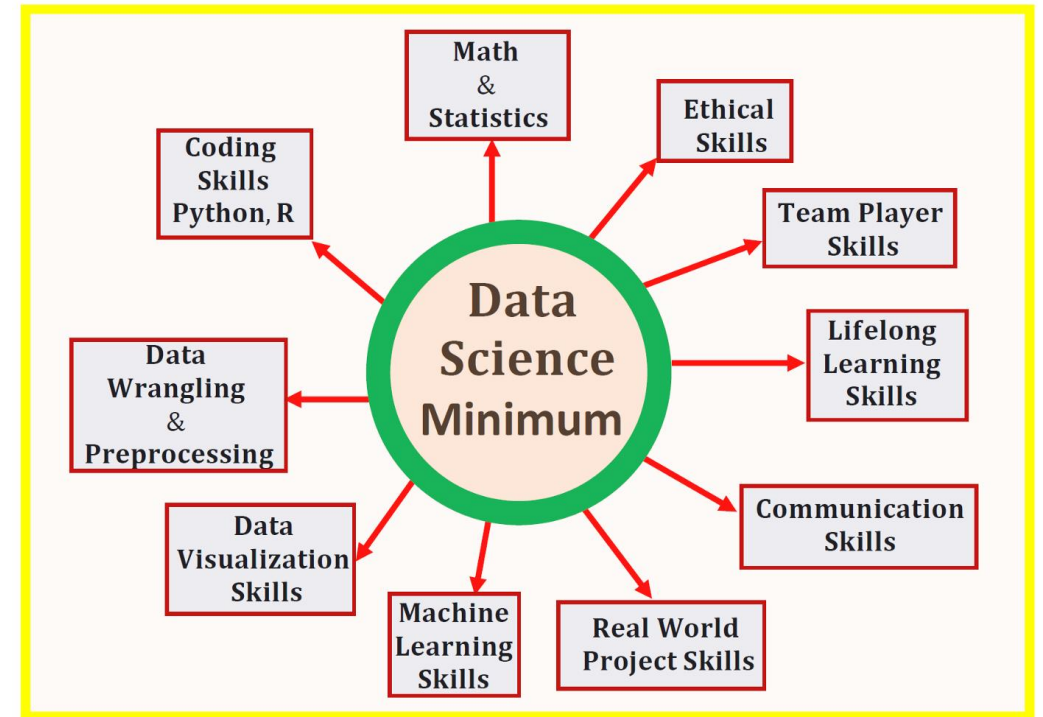


Wrap up and conclusions



Topics covered

- ~~What is Data Science?~~
- ~~Python basics~~
- ~~Descriptive statistics~~
- ~~Array computations~~
- ~~Manipulating data tables~~
- ~~Data visualization~~
- ~~Mapping~~
- ~~Text manipulation and data cleaning~~
- ~~Statistical perspective: hypothesis tests and confidence intervals~~
- ~~Machine learning perspective: supervised and unsupervised learning~~



Learning goals

1. Understand concepts in data science

- Learn basic computational skills for analyzing data
- Understand concepts in statistics and machine learning

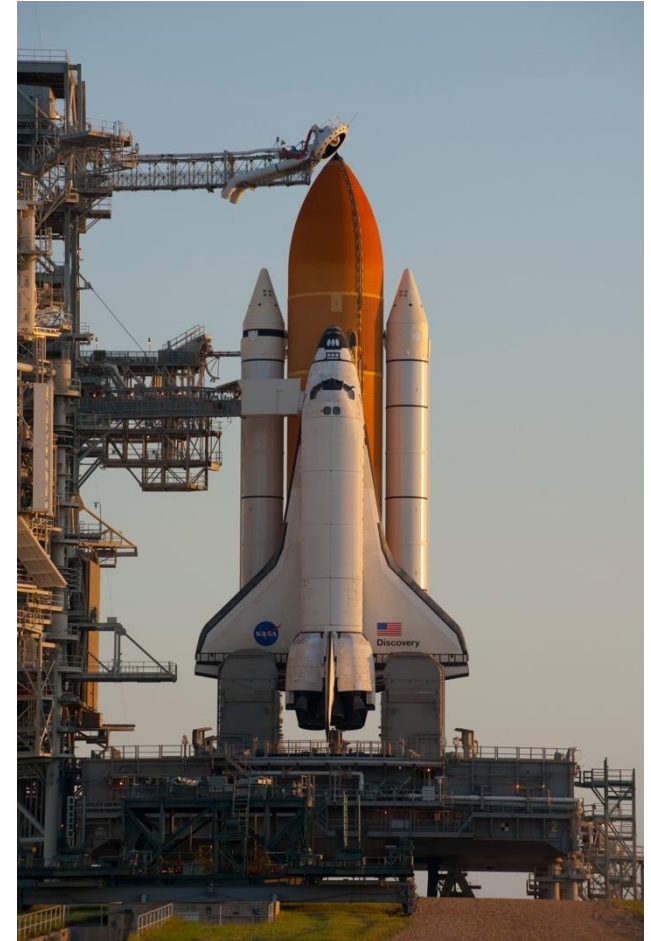
2. Gain practical data science skills applicable to any domain

3. See how data science analyses can be applied to real-world data from a variety of domains

- There will be ~weekly readings on data science related topics

There are no prerequisites for this class

- E.g., no prior knowledge of statistics or programming is required

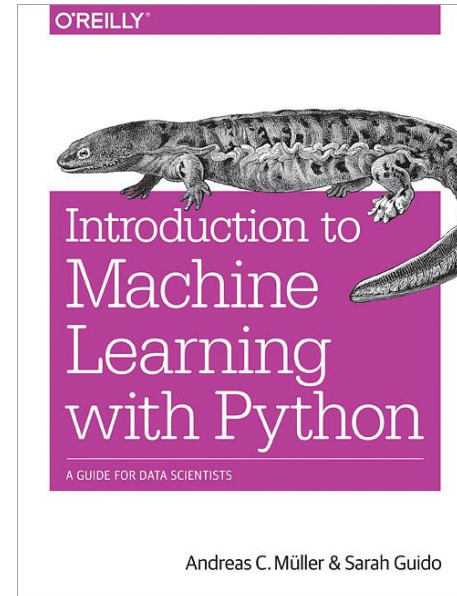
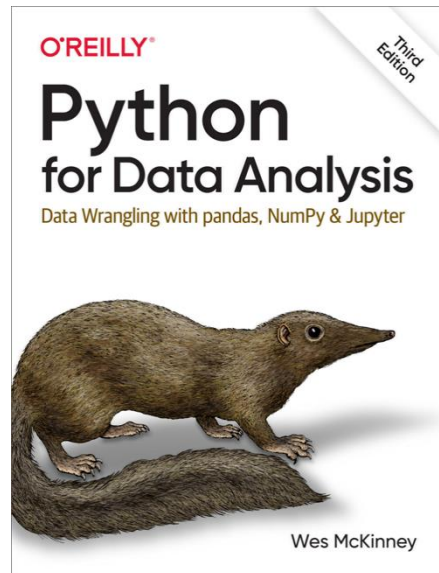


We had fun along the way???



Next steps

1. Take more advanced Statistics and Data Science classes offered at Yale!
 - S&DS 1000, S&DS 2400, YData connector classes, ...
2. There are many good books and online resources to learn more Python



Teaching Assistants

Preceptor

- Shivam Sharma

Teaching Fellows

- Yidan Mai
- Minoli Daigavane
- Ruixiao Wang

Undergraduate Learning Assistants

- Kyle Levesque
- Brunokai Ong
- Sloane Huey
- Christian Baca



Good luck with the end of the semester!



Keep in touch: yuejie.chi@yale.edu